How do demand fluctuations and credit constraints affect R&D? Evidence from Central-, Southern and Eastern Europe

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February 28, 2011

Abstract

The opportunity cost approach in growth and volatility literature suggests a countervailing cyclical effect between R&D and short-term investments. This exposition is a subject of theoretical and empirical debate. We extend the discussion by investigating the impact of demand fluctuations and credit constraints upon firm R&D in 10 new EU member states from Central-, Southern and Eastern Europe (CSEE). Overall, we find supportive evidence on the opportunity cost argument. However the country's level of advancement, as defined by the membership in OECD, matters in how the firm R&D varies with demand and credit constraints. We observe more demand driven behaviour amongst the firms operating in non-OECD CSEEs, whereas the impact of credit constraints remains weak. On the contrary, in more advanced OECD CSEEs, the firms dependedness on credit markets is higher and the credit constraints have a stronger impact upon firm R&D.

JEL Code: G31, E32, O30, O52

Keywords: R&D cyclicality; demand shocks; credit constraints

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The research contribution of Kadri Männasoo has been supported by Estonian Science Foundation grant project ETF8796.

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‡The authors thank Martti Randveer, Karsten Staehr and Dmitri Kulikov for their insightful comments. All remaining errors are authors.

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1. Introduction

The last two decades on convergence road have demonstrated remarkable growth episodes in the Central, Southern and Eastern European (CSEE) economies, but also severe downturns with a magnitude far exceeding slowdowns in mature economies. The global financial crisis has made these vulnerabilities even more evident. However the countries in CSEE region have significant diversity in terms of technological advancement and financial sector development. Six out of ten new EU member countries - Czech Republic, Hungary, Estonia, Poland, Slovakia and Slovenia belong to OECD, whereas the rest of four - Latvia, Lihtuania, Bulgaria and Romania are still "knocking on the door"'.

The modern theories on endogenous growth and the Schumpeterian growth theory in particular stress the importance of innovation, R&D and knowledge creation being vital for long-term sustainable growth. The so called cleansing mechanism or "virtue of bad times" according to the Schumpeterian view of business cycles suggests that recessions help to correct for inefficiencies whilst forcing firms to focus on productivity enhancing agenda. An opportunity cost effect steps in here: long-term innovative activities gain priority over short-term capital investments at the time of recession. The countercyclical nature of innovation is an appealing property. Evening out some of the cyclical volatility enables to navigate towards a more balanced development path. However, the counter-cyclicality argument of long-term productivity enhancing investments will be valid only to the extent that firms' access to credit is not limited. This is hardly the case whenever the firm is hit by an adverse (idiosyncratic or aggregate) shock. A negative shock has an effect on firm's current earnings, reducing the availability of internal funds as well as undermines the firms' capacity to borrow or raise external funds in general. Hence, the extent to which credit constraints bind depends on whether the firm faces an upturn or downturn on the market.

The aim of the current paper is to look how the firm likelihood to undertake R&D is affected by the fluctuating demand and credit constraints. We divide the 10 new EU member states from Central, Southern and Eastern Europe into two groups according to OECD membership status in order to see whether the countries advancement level matters in how the demand fluctuations and credit constraints impact the likelihood to conduct R&D.

The paper is organized as follows. Sections 2 and 3 provide a literature survey discussing the cyclicality of R&D and the credit constraints issues. In section 4 we present the empirical analysis and results including the subsections on methodology, data and estimations on OECD6 and Non-OECD4 countries. Section 5 concludes.

2. The cyclicality of R&D

The avenue of research investigating the impact of volatility upon growth has moved to the forefront in the research agenda (Francois and Lloyd-Ellis (2009)). The influential model proposed by Aghion, Angeletos, Banerjee, Manova - AABM (see Aghion et al. (2005, 2010)) starts with a distinction made between short-run capital investments and long-term productivity enhancing investments such as R&D. The argument is made that in the absense of credit constraints, long-run growth enhancing investments behave in a countrercyclical manner. The intuition is straightforward and pertains to opportunity cost effect. Literally taken, as lower demand makes the return on short-term investments lower in recessions, the opportunity cost of long-run productivity-enhancing investments becomes also lower. This implies that it is more profitable to invest in short-run production with high-level productivity at the time of peak (facing positive demand) than in long-term R&D, which involves an uncertainty component and delay in returns. An adverse productivity shock on the contrary motivates entrepreneurs to invest in long-term R&D as the opportunity cost in terms of forgone current production is low. The theory is appealing also from the social welfare aspect as the reallocating a proportion of R&D from booms to recessions would allow the economy to grow at a lower resource cost (Barlevy (2007)).

Aghion et al. (2010) however claim that under sufficiently tight credit constraints the long-term investments become procyclical leading to both "‘lower mean growth and amplified volatility". Their model identifies a propagating mechanism that credit constraints have upon cyclical composition of investment. In particular, there exists a wedge, which reflects the probability of failure determined by allocation of investments between short- and long-term. This wedge is counter-cyclical, decreasing in booms and increasing in recessions. The intuition is that a negative shock will leave firms short of internal ressources, as well as limit access to external funds. Hence, the probability of failure increases and the shift from the long-term to the short-term investments simply improves the liquitity available over the next period.

Barlevy (2007) proposes an opposing theory claiming that R&D expenditures behave in a pro-cyclical manner contrary to the opportunity cost argument. According to his model R&D will be more biased toward boom "the bigger the ratio of profits in booms to their value in recessions". This to be true requires that profits are more pro-cyclical than the cost of R&D. He also shows that industries with more pro-cyclical profits proxied by stock values tend to have more procyclical R&D growth. A number of references to earlier literature are given in Barlevy (2007), which confirm pro-cyclical behavior of R&D.

Indeed, the empirical evidence has not given consistent support to the R&D counter- or pro-cyclicality argument. Studying long-run relationships in U.S. aggregate data Rafferty (2003) demonstrates pro-cyclical and asymmetric patterns of firm-financed R&D. He claims that increased after-tax cash flows rise R&D expenditures and that more R&D has been lost during recessions than gained during expansions with negative net effect of business cycles on aggregate R&D. The evidence on an annual panel of 20 U.S. manufacturing industries over 1958-1998 (Ouyang (2007)) indicates a more mixed result with pro-cyclical R&D responding asymmetrically and negatively to demand shocks. In the follow-up paper by Ouyang (2010) she constructs a model which suggests that R&D appears counter-cyclical with low credit constraints, but procyclical with sufficiently high credit constraints, whereas mixed cyclicality is displayed for moderate degree of credit constraints. Her theory also claims that Schumpeterian "virtue of bad times" holds only if "the marginal opportunity cost of R&D dominates over the marginal expected return".

Aghion et al. (2005, 2010) provide also the empirical proof to their model. Based on an annual panel of 21 OECD countries over the 1960-2000 period, the analysis shows that long-term growth-enhancing investments respond less to positive exogenous shocks in countries with more developed financial sectors. The evidence on AABM model is however limited. Aghion et al. (2008) have given also firm-level evidence to their model using a panel dataset covering 13 000 French firms over the period of 1980-2000. They use a proxy variable called "payment incident" (a record of payment failure in a blacklist, which affects firms' access to new credit) in order to measure credit contraints. AABM show a stronger positive correlation between sales and R&D spending in more credit-constrained firms. Also the credit-constrained firms suffer more from demand volatility, having an asymmetric effect on R&D investments, which become more harmed in slumps than they are encouraged in booms. In similar line Bovha-Padilla et al. (2009) conduct panel study on Slovenian firms for the period 1996-2002 and observe the pro-cyclicality of R&D investment in credit constrained firms, whereas the effect disappears in less financially dependent firms, which have access to parent company funding or government subsidies. A recent contribution by Voigt and Moncada-Paterno-Castello (2009) suggests that corporate R&D in EU has remained resilient to the economic and financial crisis in 2008-2009.

3. Credit frictions

Theoretical literature on growth, business cycles and firm investment behavior is increasingly more concerned about imperfect capital markets. The asymmetric information problem, uncertain and lagged returns make R&D investments particularly sensitive to credit constraints.[1]

The impact of credit constraints on firm performance is predominantly negative. A recent evidence by Campello et al. (2010) on the global financial crisis of 2008/09 shows that constrained firms in U.S., Europe, and Asia witnessed deeper cuts in employment, technology and capital spending. Their study also interestingly points to the issue that constrained firms drew more heavily on lines of credit in order to frontload funds in fear of restricted access to credit in the future. Savignac (2008), Aghion et al. (2008), Ouyang (2007, 2010) and others have found strong support to the evidence that financial and credit constraints have an adverse effect upon R&D and innovation. Badia and Slootmaekers (2008) study on the relationship between productivity and financial constraints in Estonia concluded that except of all other industries financial constraints had a large negative impact on productivity in R&D sector.

A reverse relationship or the impact of R&D intensity upon liquidity constraints is tested by Piga and Atzeni (2007). Their empirical findings based on a survey of Italian manufacturing firms show that credit constraints depend on the R&D intensity of the firm and that an inverse U-shaped relationship is observed between R&D activity and the probability to be credit constrained. They also note that firms with no R&D are less likely to apply for new credit.

The measures of credit constraints vary across the studies. Unavailability of explicit information imposes that indirect measures of financial constraints such as firm size, age, dividends distribution, credit rating etc prevail in the literature. One typical approach to go about financial constraints is the measurement of investments sensitivity to internally generated cash flows using the Q-theory and Euler-equation models. A recent survey on financial constraints measurement issues is provided by Hadlock and Pierce (2010) where they also propose their own novel approach. Hadlock and Pierce (2010) find that firms' age and size alone perform as good predictors of the level of financial constraint.

In empirical front Ouyang (2007, 2010) employs two proxies capturing financial constraints faced by U.S. manufacturing industries - firm liquid assets and its net worth. In her interpretation the first variable reflects the firm's need for external funds, whereas the net worth acts as the collateral for a loan. As mentioned above, Aghion et al. (2008) use a "payment incident" or blacklist record as a proxy variable for picking up credit constraints on firm-level.

[1]The adverse selection between investors financing R&D and entrepreneurs undertaking R&D has been investigated by Plehn-Dujowich (2009) showing that an increase in the mean skill level enhances growth via greater R&D productivity and investment.

Campello et al. (2010) however argue in favor of a direct survey-based measure of financial constraint demonstrating that traditional constraint measures fail to identify meaningful patterns in their sample survey data. In the same vein, Kaplan and Zingales (1997) question the measurement of financial constraints via investment-cash-flow sensitivities, extracting access to credit information from firms public statements instead. Empirical evidence on the use of direct financial or credit constraint measures is scarce due to limited data. Ayyagari et al. (2008) analysis draws on the World Bank Business Environment Survey [2], where they capture firm managers' direct response to perceived financial obstacles. Also Savignac (2008) employs the direct, qualitative indicator on financial constraints derived from the survey conducted by the French Ministry of Industry in order to obtain information about the financing conditions of innovative projects of manufacturing firms in France. His arguments in favor of a direct measure on financial constraints are that it avoids interpretation problems of indirect indicators, such as cash-flows and that it provides specific (and new) information about the financial problems encountered by firms, whereas accounting variables or credit rating index reflect the global financial situation of the firm Savignac (2008).

4. Empirical analysis and results

4.1. Methodology

Our econometric approach departs from recursive bivariate probit model. According to Monfardini and Radice (2008) the bivariate probit model with endogenous dummy is the appropriate inference tool "whenever there are good "a priori" reasons to consider a dependent binary variable to be simultaneously determined with a dichotomous regressor"'.

Savignac (2008) has employed a recursive bivariate probit for estimating the French firms propensity to innovate being subject to endogenous financial constraints. Masso and Vahter (2008), Masso et al. (2010), employ a bivariate probit model for estimating the knowledge production function in respect to Estonian firms' product and process innovation used in later modeling stages for investigating the linkages between productivity and innovation and the FDI impact upon innovation respectively. The credit rationing patterns of R&D intensive firms have been studied with a bivariate probit model by Piga and Atzeni (2007).

In our model the endogenous financial constraint is regressed with the fol-

[2]The Business Environment survey resembles the Business Environment and Enterprise Performance Survey (BEEPs), the datasource for our our empirical analysis.

lowing variables: (1) log of firm age in years since starting operation in particular country; (2) firm size measured by number of employees; (3) a dummy variable reflecting publicly listed firms; (4) the share of foreign ownership; (5) private bank funding in firm total fixed investments funding, (6) a dummy variable on the presence of 90-day overdue loans; (7) the share of sales sold on credit; (8) an indicator on whether the firm is audited and finally (9) the dummy variable on the existence of state subsidies. [3]

The argument in favor of a recursive model is that financial constraints can be considered endogenous to R&D. Not only are the financial constraints making impact upon the firm likelihood to conduct R&D, but also the qualities which distinguish R&D engaging firms such as skill and technology intensity or competitiveness turn them more attractive for creditors. Pertaining to the above estimating separately the likelihood to conduct R&D and likelihood to be financially constrained would lead to inconsistent results. A two-step procedure where predicted values from financial constraint equation (a selection equation) are fed into the R&D equation (outcome equation), is potentially inefficient insofar as it does not account for the possible correlation between the disturbance terms in the two equations**?**. Binary models in general are demanding in terms of sample sizes, moreover so in bivariate binary outcome models (Monfardini and Radice (2008)).

Considering a recursive system with binary endogenous variables we get:

$$\begin{cases} y_1 = \beta_1 x_1 + \epsilon_1 \\ y_2 = \beta_2 x_2 + \gamma_2 y_1 + \epsilon_2 \end{cases}$$

Where y_1 represents the unobserved severity of financial constraints in a reduced form equation and y_2 stands for the firm likelihood to conduct R&D in the structural form equation. x_1 and x_1 denote the exogenous variables explaining presence of financial constraints and the R&D decision respectively. The errors ϵ_1 and ϵ_2 are jointly normally distributed with zero mean, unit variance and correlation of ρ where $|\rho| > 0$.[4] The correlation between error terms can be interpreted as the correlation between the unobservable explanatory variables of the two equations.

A widespread opinion in the literature is that the parameters of the second equation in structural form are not identified unless the reduced form equation contains at least one variable not part of the regressors in structural form equa-

[3] In comparison Savignac (2008) estimates the firm financial constraints using the five measures listed below: (1) the share of the banking debt, (2) the share of the own financing in the firm's total financing resources, (3) logarithm of tangible assets as a proxy for the collateral, (4) firm gross operating profit margin ratio and finally (5) the firm size.

[4] In case $\rho = 0$ two separate probit models can be estimated implying that y_1 is exogenous for the structural form equation.

tion. This assertion, stated by Maddala (1983) is contradicted in a recent paper by Wilde (2000), who show that exclusion restrictions are not needed provided there is one varying exogenous regressor in each equation.Monfardini and Radice (2008)

For MLE four probabilities (summing up to 1) are needed like in a standard bivariate probit model without endogenity as follows Lee (2010):

$$\begin{aligned}
Pr(y_1=1,y_2=1) &= P(\varepsilon_1 > -\beta_1 x_1, \varepsilon_2 > -\gamma_2 - \beta_2 x_2)\\
Pr(y_1=1,y_2=0) &= P(\varepsilon_1 > -\beta_1 x_1, \varepsilon_2 < -\gamma_2 - \beta_2 x_2)\\
Pr(y_1=0,y_2=1) &= P(\varepsilon_1 < -\beta_1 x_1, \varepsilon_2 > -\beta_2 x_2)\\
Pr(y_1=0,y_2=0) &= P(\varepsilon_1 < -\beta_1 x_1, \varepsilon_2 < -\beta_2 x_2)
\end{aligned}$$

As y_1 and y_2 are observed as dichotomous variables, it is necessary to adopt the standard normalization of the variance of the errors. Given $\sigma_1 = SD(\varepsilon_1)$ and $\sigma_2 = SD(\varepsilon_2)$ the respective standardized probabilities are obtained as functions of $\beta_1/\sigma_1, \gamma_1/\sigma_1, \beta_2/\sigma_2, \rho$ where the last term ρ denotes correlation between standardized error terms.

$$\begin{aligned}
Pr(-\tfrac{\varepsilon_1}{\sigma_1} < \tfrac{\beta_1}{\sigma_1}x_1, -\tfrac{\varepsilon_2}{\sigma_2} < \tfrac{\gamma_2}{\sigma_2} + \tfrac{\beta_2}{\sigma_2}x_2) &= \Psi(\tfrac{\beta_1}{\sigma_1}x_1, \tfrac{\gamma_2}{\sigma_2} + \tfrac{\beta_2}{\sigma_2}x_2; \rho)\\
Pr(-\tfrac{\varepsilon_1}{\sigma_1} < \tfrac{\beta_1}{\sigma_1}x_1, \tfrac{\varepsilon_2}{\sigma_2} < -\tfrac{\gamma_2}{\sigma_2} - \tfrac{\beta_2}{\sigma_2}x_2) &= \Psi(\tfrac{\beta_1}{\sigma_1}x_1, \tfrac{-\gamma_2}{\sigma_2} - \tfrac{\beta_2}{\sigma_2}x_2; -\rho)\\
Pr(\tfrac{\varepsilon_1}{\sigma_1} < -\tfrac{\beta_1}{\sigma_1}x_1, -\tfrac{\varepsilon_2}{\sigma_2} < \tfrac{\beta_2}{\sigma_2}x_2) &= \Psi(-\tfrac{\beta_1}{\sigma_1}x_1, \tfrac{\beta_2}{\sigma_2}x_2; -\rho)\\
Pr(\tfrac{\varepsilon_1}{\sigma_1} < -\tfrac{\beta_1}{\sigma_1}x_1, \tfrac{\varepsilon_2}{\sigma_2} < -\tfrac{\beta_2}{\sigma_2}x_2) &= \Psi(-\tfrac{\beta_1}{\sigma_1}x_1, -\tfrac{\beta_2}{\sigma_2}x_2; \rho)
\end{aligned}$$

From here the maximum likelihood is derived by maximizing:

$$\sum[y_{1i}y_{2i}ln(\Psi(\tfrac{\beta_1}{\sigma_1}x_1, \tfrac{\gamma_2}{\sigma_2}+\tfrac{\beta_2}{\sigma_2}x_2;\rho)+y_{1i}(1-y_{2i})ln(\Psi(\tfrac{\beta_1}{\sigma_1}x_1, \tfrac{-\gamma_2}{\sigma_2}-\tfrac{\beta_2}{\sigma_2}x_2;-\rho)$$
$$+(1-y_{1i})y_{2i}ln\Psi(-\tfrac{\beta_1}{\sigma_1}x_1, \tfrac{\beta_2}{\sigma_2}x_2;-\rho)+(1-y_{1i})1-y_{2i})ln\Psi(-\tfrac{\beta_1}{\sigma_1}x_1, -\tfrac{\beta_2}{\sigma_2}x_2;\rho)]$$

4.2. Data

This paper employs the data from Business Environment and Enterprise Performance survey (BEEPs) conducted jointly by EBRD and World Bank. Three subsequent rounds, 2002, 2005 and 2009 of BEEPs have been employed. The survey covers firm-level data from a wide set of transition countries, however to reduce the heterogeneity of the sample the 10 new EU member countries: Czech Republic, Bulgaria, Hungary, Poland, Slovakia, Slovenia, Romania, Estonia, Latvia and Lithuania, have been selected for the empirical analysis.

The sample structure of BEEPs survey has been designed to be representative to the population of firms [5] The exclusions are made regarding firms

[5] See BEEPs reports on methodology and observations at http://www.ebrd.com/pages/research/analysis/surveys/beeps.shtml for more details on survey design.

operating in sectors under government regulation and prudential supervision such as banking, electric power, rail transport and water supply. Also the firms with only one employee or with a number of employees in excess of 10 000 were excluded. Similarly, firms with yearly sales below 50 000 euros (unlikely to conduct R&D) and firms with less than three years in operation were eliminated [6].

Table 1: Variables Description

NAME	UNIT	DESCRIPTION
RD	[0;1]	1 if firm conducts R&D, 0 otherwise
constrained	[0;1]	1 if firm is constrained, 0 otherwise
age	ln(year)	age in years since company started operations in particular country. For transition countries the beginning year is set to 1987 if reported earlier
size	[0;1]	dummy variable on wheter the company belongs to one of the three size categories: 1-49 employees; 50-250 employees or 250-10 000 employees
dsales	%	Percent change in sales over last three years in real terms
UniGrade	%	A percent of firm workforce having university degree or higher
ExSale	%	share of direct and indirect exports in firm total sales
BankFin	%	Proportion of fixed assets (land, buildings, machinery, equipment) financed with private bank borrowing
CredSale	%	Proportion of sales sold on credit i.e. paid after delivery
foreign	%	Percent of foreign ownership if foreign share $\geq$50%, zero otherwise
Overdue	[0;1]	1 if the firm has 90 day overdue payment (includes tax overdues and overdues on utilities), 0 otherwise
audit	[0;1]	1 if the financial statements reviewed by external auditor, 0 otherwise
subsidies	[0;1]	1- if the firm has been subject to public subsidies from local, national or EU sources
VA	%	Industry level growth in real value added
LAB	%	Industry level annual growth of workforce
TURN	%	Industry level real annual sales growth

The BEEPs survey contains valuable information about the firm-specific factors, including firm age, size measured by the number of employees, ownership, sales growth, share of export, employees skills level as well as the dependedness on and access to external finances. The data relevant for the current study refer to the years 2001, 2004 and 2007. See for the description

[6]Starting businesses might exhibit dynamics, which are not well in line with general patterns on firm or industry level

of the variables from Table 1 above.

The credit constraint variable is conditioned on two terms, first the firm dependedness on external finance and secondly the access to finance. The firms which state that they don't need a loan are defined as not dependent on external finance, whereas the access to finance is irrelevant for them. On the other hand these firms which do not have a loan because they claim not to be eligible for applying one can be treated as discouraged and hence credit constrained. In addition to discouraged firms the firms which have been applying for credit, but turned down by the bank are attached to the credit constrained group. See the Table 2 below.

Table 2: Variable definitions

	DEPENDENT	INDEPENDENT
CONSTRAINED	Loan rejected OR discouraged from applying a loan	Not applicable
UNCONSTRAINED	Has got a loan	Does not need a loan

The demand shock has been proxied by three industry level variables i.e. year-to-year growth in real value added, employment and real turnover. Asymmetric effects are accounted for by decomposing the demand variables into separate variables for positive values i.e. growth and negative valuse i.e. decline.

$VA^{+} = \Delta VA$ if $\Delta VA > 0, \quad 0$ otherwise
$VA^{-} = \Delta VA$ if $\Delta VA < 0, \quad 0$ otherwise

See for the full set of variables used in the current analysis from the data description table, Table 1, above .

4.3. R&D in more and less advanced CSEECs

The CSEE countries have witnessed severe episodes of volatility over the past two decades in struggle for improved international competitiveness. The econometric results reveal exciting cyclical patterns reflecting how R&D reacts to demand shock and to which extent do the credit constraints matter in firm R&D decisions.

The estimation outcome (see Table 3) implies that credit constraints curb firms R&D initiative only in CSEECs belonging to OECD. Negative coefficients appear in all regressions with two being highly statistically significant.

The contrary happens for the non-OECD4, where all signs for the credit constraint variable are positive, whereas one of them significant at 10% level of statistical significance. This result seems to be counterintuitive while anticipating that credit constraints should be a bigger issue for non-OECD sample. However, the credit constraint variable captures not only restriced access to credit, but also the firms dependence on external finance. The share of credit dependent firms amongst the OECD members turns to be more than twice the same share in non-OECD sample, 45% and 20% respectively (See the descriptive statistics regarding the OECD6 and Non-OECD4 countries from **??**). Hence, the firms operating in countries with more developed credit markets become more reliant on external financing, having possibly also higher expectations in regard to credit availability. Table 7 demonstrates the results for credit dependent firms - showing that for both country groups the relationship between credit constraints and R&D becomes negative, though due to a small sample size few statistically significant signs appear.

An important R&D accelerating factor is firm sales or capacity to generate internal funds. Internal funds reduce the firm need to borrow or make it even redundant. The stronger the firm revenue generating power the more likely it is engaged in R&D. This result is valid for both country groups, reassuring that sales growth remains important despite the country's development level or advancement in credit markets. Similar to firm sales growth the size measured by the number of full-time employees is positively and significantly related with R&D across all new EU member countries from CSEE region not depending on their level of advancement.

In OECD6 the firms having higher share of exports as well as the listed firms tend to be more likely to pursue R&D. The share of university degreee employees has a modest impact upon R&D, being significant for financially dependent firms in OECD6 only. Seemingly the education level has strongest implications in a sample of firms located in more advanced markets and firms which are financed by external creditors. Interestingly the company age and foreign ownership elicit no significant effect upon firm likelihood to conduct R&D.

The industry level demand fluctuations have a significant impact upon R&D. Whilst the firm own sales behaves in a pro-cyclical manner, the sector demand indicators show more complex patterns. Decreasing industry-level demand is counter-cyclical with R&D providing support to the opportunity cost theory. Hence, the "virtue of bad times"' proves to be true in CSEE countries being valid for both the OECD member countries as well as for the non-members. On the positive side of demand shock the evidence is rather modest with mixed signs across the three demand variables. Although, the non-OECD members seem to behave more counter-cyclically on both strands of the demand cycle. Hence, the estimations reveal more counter-cyclicality amongst the non-OECD members, suggesting that the opportunity cost argumentation is more valid for firms operating in less advanced environment.

Ouyang (2010) theory might provide the explanation. She suggests that there is a trade-off between marginal opportunity cost and marginal expected return from R&D. Arguably the closer the country to the techonogical frontier the higher are the expected returns from R&D relative to the opportunity costs. This supposition however remains a subject for further research.

Table 3: R&D and credit constraints in OECD6 versus non-OECD4, 2002-07, Total sample

R&D=1	OECD6			Non-OECD4		
Industry demand proxy	Value added	Employment	Turnover	Value added	Employment	Turnover
constrained (d)	-0.209***	-0.201***	-0.145	0.362	0.242	0.456*
	(0.061)	(0.071)	(0.110)	(0.417)	(0.766)	(0.265)
lnage	-0.010	-0.013	-0.011	-0.001	0.019	0.009
	(0.027)	(0.028)	(0.027)	(0.026)	(0.027)	(0.025)
empl2to49 (d)	-0.281***	-0.286***	-0.301***	-0.153***	-0.140**	-0.143***
	(0.041)	(0.041)	(0.037)	(0.048)	(0.062)	(0.042)
empl50to250 (d)	-0.117***	-0.120***	-0.114***	-0.060**	-0.051	-0.054*
	(0.024)	(0.024)	(0.023)	(0.030)	(0.038)	(0.028)
listed (d)	0.142***	0.148***	0.143***	0.044	0.051	0.077
	(0.043)	(0.044)	(0.047)	(0.046)	(0.050)	(0.049)
ExSale	0.088**	0.091**	0.093***	-0.020	-0.028	-0.014
	(0.036)	(0.036)	(0.033)	(0.034)	(0.036)	(0.033)
foreign	-0.025	-0.019	-0.010	0.030	0.030	0.033
	(0.031)	(0.031)	(0.030)	(0.031)	(0.033)	(0.030)
UniGrade	0.030	0.020	0.033	0.058	0.056	0.071*
	(0.036)	(0.036)	(0.035)	(0.038)	(0.042)	(0.037)
dsales	0.109***	0.106***	0.102***	0.070***	0.086***	0.071***
	(0.025)	(0.025)	(0.025)	(0.023)	(0.028)	(0.022)
Dneg	-1.927***	-0.790**	-2.670***	-4.563	-2.112**	-0.624***
	(0.614)	(0.329)	(0.845)	(3.872)	(0.901)	(0.192)
Dpos	-0.149	0.199	0.111	-1.036***	-0.995**	-0.096
	(0.217)	(0.319)	(0.102)	(0.352)	(0.427)	(0.160)
lnage	-0.044***	-0.044***	-0.047***	-0.031	-0.031	-0.033
	(0.016)	(0.016)	(0.017)	(0.021)	(0.022)	(0.021)
empl2to49 (d)	0.017	0.018	0.026	-0.030	-0.033	-0.029
	(0.029)	(0.029)	(0.029)	(0.033)	(0.034)	(0.032)
empl50to250 (d)	-0.003	-0.002	0.003	-0.028	-0.029	-0.029
	(0.029)	(0.029)	(0.030)	(0.027)	(0.028)	(0.027)
listed (d)	0.094**	0.094**	0.093**	-0.015	-0.015	-0.018
	(0.039)	(0.039)	(0.040)	(0.037)	(0.038)	(0.037)
foreign	-0.055**	-0.057**	-0.057**	-0.031	-0.031	-0.030
	(0.023)	(0.023)	(0.023)	(0.029)	(0.030)	(0.028)
BankFin	-0.058**	-0.060**	-0.067**	-0.067*	-0.068	-0.064*
	(0.029)	(0.029)	(0.028)	(0.036)	(0.044)	(0.035)
overdue (d)	0.123***	0.126***	0.140***	0.142***	0.142***	0.135***
	(0.042)	(0.042)	(0.040)	(0.041)	(0.042)	(0.041)
CredSale	-0.037**	-0.034**	-0.037**	-0.025	-0.027	-0.024
	(0.015)	(0.016)	(0.016)	(0.023)	(0.024)	(0.022)
audit (d)	-0.021	-0.021	-0.017	0.007	0.004	0.010
	(0.014)	(0.014)	(0.016)	(0.021)	(0.028)	(0.019)
subsidies (d)	-0.063***	-0.063***	-0.063***	-0.066***	-0.067***	-0.063***
	(0.014)	(0.014)	(0.015)	(0.023)	(0.024)	(0.024)
athrho	0.836**	0.791*	0.500	-0.498	-0.321	-0.658
Log likelihood	-2698.70	-2702.78	-2700.90	-1234.34	-1226.62	-1235.28
No of obs.	3394	3394	3394	1619	1619	1619

Source: authors' calculations on BEEPs data. Note: Bivariate probit marginal effects: direct effects on R&D above and indirect effects via credit constraint below. Robust, sector clustered standard errors in parenthesis. Country, sector and year dummies included. ***, **, * stand for 1%, 5% and 10% level statistical significance respectively.

5. Summary

In our study we have seeked to fill some of the gap between macroeconomic understanding of volatility and long-term growth on the one hand and the firm-level evidence of productivity enhancing R&D on the other hand. The analysis has provided solid support to the existing literature, highlighting the link between short-run demand fluctuations and long-run growth fostered by productivity enhancing R&D. The firms in CSEE countries proved to be more inclined to conduct R&D at times of low demand. This evidence is in line with opportunity cost arguments suggesting that recessions force firms to focus on productivity enhancing agenda. The comparison of CSEEs belonging to OECD versus non-members reveals that the opportunity cost driven behavior is more evident for countries with lower level of advancement.

Theory proposed by Ouyang (2010) might provide the explanation. Her model suggests that there is a trade-off between marginal opportunity cost and marginal expected return from R&D. Our interpetation is that more advanced economies being closer to the techonogical frontier are more likely to have higher expected return from R&D relative to opportunity costs. This supposition however is a subject for further research.

The financial frictions have been accounted for by a simultaneous estimation procedure - a recursive probit model on firm R&D and credit constraints. The impact of credit constraints upon R&D turned to be important for the more advanced CSEECs being the members of OECD. Moreover the firm sales growth has demonstrated a strong, significant impact in favor of R&D in most estimations. Hence, the firm revenue generating power - sales growth - reduces the financial constraints. Also many other intutive results appeared proving that the firm size decreases the exposure to credit constraints and supports R&D in both country groups, whereas the openness measured by the share of exports counts for more advanced countries only. The employees skills level proxied by the share of highly educated employees has a significant positive effect upon R&D in credit dependent firms operating in more advanced countries.

The overall implications suggest that the country's level of advancement has a significant role in how the firms R&D reacts upon the changing demand and credit constraints. The demand fluctuations seem to be more relevant for the less developed countries, whereas the credit constraints have a larger impact for the firms operating in more advanced markets.

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6. Appendixes

Table 4: Summary statistics

Variable	OECD6			Non-OECD4		
	Mean	Std.Dev.	N	Mean	Std.Dev.	N
RD	0.213	0.41	4534	0.195	0.396	2146
constrained	0.13	0.336	4534	0.12	0.325	2146
age	12.747	4.569	4534	11.864	4.381	2146
empl2to49	0.698	0.459	4531	0.632	0.482	2144
empl50to250	0.216	0.411	4531	0.262	0.44	2144
empl250to10000	0.086	0.281	4531	0.105	0.307	2144
dsales	0.134	0.372	4534	0.217	0.462	2146
UniGrade	0.172	0.241	4439	0.215	0.258	2084
listed	0.053	0.224	4534	0.047	0.211	2146
ExSale	0.142	0.272	4529	0.15	0.306	2145
ForOwned	0.129	0.335	4534	0.141	0.348	2146
ForCapShare	0.121	0.308	4523	0.128	0.31	2140
BankFin	0.121	0.265	3485	0.171	0.319	1668
overdue	0.067	0.25	4534	0.078	0.269	2146
CredSale	0.509	0.413	4505	0.477	0.397	2139
audit	0.53	0.499	4534	0.474	0.499	2146
subsidies	0.136	0.343	4534	0.074	0.262	2146
dsales	0.134	0.372	4534	0.217	0.462	2146
GDP	0.06	0.053	4534	0.095	0.05	2146
GDP^+	0.073	0.042	3991	0.096	0.05	2141
GDP^-	-0.033	0.029	543	-0.035	0.013	5
LAB	0.02	0.053	4534	0.043	0.066	2146
LAB^+	0.045	0.042	2974	0.06	0.057	1754
LAB^-	-0.028	0.038	1535	-0.036	0.043	392
TURN	0.103	0.138	4534	0.085	0.108	2015
$TURN^+$	0.118	0.138	4039	0.11	0.079	1817
$TURN^-$	-0.022	0.021	495	-0.142	0.062	198

Source: authors’ calculations on BEEPs data.
Note: OECD6 includes Czech Republic, Estonia, Hungary, Poland, Slovakia and Slovenia. Non-OECD4 includes Latvia, Lithuania, Bulgaria and Romania.

Table 5: Cross-correlation table OECD6

Variables	RD	constrained	dsales	VA	VA^{+}	VA^{-}	LAB	LAB^{+}	LAB^{-}	TURN	$TURN^{+}$	$TURN^{-}$
RD	1.000											
constrained	-0.014	1.000										
	(0.346)											
dsales	0.143	-0.108	1.000									
	(0.000)	(0.000)										
GDP	0.017	-0.026	0.069	1.000								
	(0.246)	(0.082)	(0.000)									
GDP^{+}	0.028	-0.015	0.070	0.966	1.000							
	(0.064)	(0.327)	(0.000)	(0.000)								
GDP^{-}	-0.023	-0.048	0.032	0.602	0.376	1.000						
	(0.114)	(0.001)	(0.031)	(0.000)	(0.000)							
LAB	-0.005	0.033	0.078	0.134	0.188	-0.101	1.000					
	(0.751)	(0.024)	(0.000)	(0.000)	(0.000)	(0.000)						
LAB^{+}	-0.002	0.034	0.073	0.081	0.143	-0.152	0.884	1.000				
	(0.875)	(0.022)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)					
LAB^{-}	-0.006	0.016	0.047	0.150	0.165	0.027	0.691	0.273	1.000			
	(0.683)	(0.281)	(0.001)	(0.000)	(0.000)	(0.071)	(0.000)	(0.000)				
TURN	0.011	0.046	0.073	0.263	0.261	0.138	0.151	0.105	0.147	1.000		
	(0.473)	(0.002)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)			
$TURN^{+}$	0.012	0.046	0.074	0.250	0.250	0.123	0.155	0.107	0.152	0.998	1.000	
	(0.404)	(0.002)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)		
$TURN^{-}$	-0.022	0.007	-0.009	0.248	0.210	0.243	-0.019	-0.003	-0.035	0.260	0.193	1.000
	(0.140)	(0.615)	(0.565)	(0.000)	(0.000)	(0.000)	(0.194)	(0.839)	(0.018)	(0.000)	(0.000)	

Source: authors' calculations on BEEPs data.
Note: OECD6 includes Czech Republic, Estonia, Hungary, Poland, Slovakia and Slovenia.

Table 6: Cross-correlation table Non-OECD4

Variables	RD	constrained	dsales	VA	VA^+	VA^-	LAB	LAB^+	LAB^-	TURN	$TURN^+$	$TURN^-$
RD	1.000											
constrained	0.046	1.000										
	(0.035)											
dsales	0.088	-0.065	1.000									
	(0.000)	(0.003)										
GDP	-0.064	-0.034	0.093	1.000								
	(0.003)	(0.118)	(0.000)									
GDP^+	-0.063	-0.033	0.093	0.999	1.000							
	(0.004)	(0.131)	(0.000)	(0.000)								
GDP^-	-0.050	-0.036	0.011	0.123	0.087	1.000						
	(0.021)	(0.099)	(0.608)	(0.000)	(0.000)							
LAB	-0.113	-0.044	0.076	0.420	0.417	0.119	1.000					
	(0.000)	(0.042)	(0.000)	(0.000)	(0.000)	(0.000)						
LAB^+	-0.087	-0.047	0.085	0.417	0.417	0.040	0.942	1.000				
	(0.000)	(0.031)	(0.000)	(0.000)	(0.000)	(0.065)	(0.000)					
LAB^-	-0.112	-0.012	0.011	0.181	0.173	0.245	0.558	0.248	1.000			
	(0.000)	(0.586)	(0.623)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)				
TURN	-0.045	-0.058	0.049	0.498	0.498	0.054	0.313	0.383	-0.036	1.000		
	(0.044)	(0.010)	(0.029)	(0.000)	(0.000)	(0.016)	(0.000)	(0.000)	(0.106)			
$TURN^+$	-0.013	-0.046	0.034	0.475	0.475	0.052	0.293	0.369	-0.069	0.916	1.000	
	(0.537)	(0.032)	(0.114)	(0.000)	(0.000)	(0.016)	(0.000)	(0.000)	(0.001)	(0.000)		
$TURN^-$	-0.104	-0.025	0.002	0.222	0.222	0.024	0.166	0.190	0.010	0.709	0.328	1.000
	(0.000)	(0.251)	(0.912)	(0.000)	(0.000)	(0.264)	(0.000)	(0.000)	(0.658)	(0.000)	(0.000)	

Source: authors' calculations on BEEPs data.
Note: Non-OECD4 includes Latvia, Lithuania, Bulgaria and Romania.

Table 7: R&D and credit constraints in OECD6 versus non-OECD4, 2002-07, financially dependent firms

R&D=1	OECD			Non-OECD		
Industry demand proxy	Value added	Employment	Turnover	Value added	Employment	Turnover
constrained (d)	-0.157**	-0.135	-0.092	-0.107	-0.115	-0.063
	(0.079)	(0.086)	(0.087)	(0.095)	(0.077)	(0.121)
lnage	0.006	0.004	0.004	-0.048	-0.025	-0.025
	(0.035)	(0.035)	(0.032)	(0.043)	(0.044)	(0.043)
empl2to49 (d)	-0.276***	-0.280***	-0.290***	-0.190***	-0.174***	-0.183***
	(0.042)	(0.041)	(0.040)	(0.049)	(0.050)	(0.050)
empl50to250 (d)	-0.109***	-0.111***	-0.107***	-0.089**	-0.075*	-0.075*
	(0.031)	(0.030)	(0.029)	(0.040)	(0.041)	(0.040)
listed (d)	0.172***	0.177***	0.176***	0.012	0.004	0.052
	(0.053)	(0.053)	(0.054)	(0.066)	(0.067)	(0.070)
ExSale	0.098**	0.100**	0.100**	0.012	-0.004	0.014
	(0.043)	(0.042)	(0.041)	(0.049)	(0.050)	(0.049)
foreign	0.000	0.011	0.015	0.031	0.032	0.035
	(0.039)	(0.038)	(0.037)	(0.049)	(0.049)	(0.047)
UniGrade	0.118**	0.102**	0.112**	0.069	0.054	0.095
	(0.049)	(0.049)	(0.047)	(0.068)	(0.069)	(0.064)
dsales	0.102***	0.095***	0.089***	0.018	0.032	0.032
	(0.032)	(0.031)	(0.031)	(0.030)	(0.030)	(0.030)
Dneg	-3.046***	-0.545	-3.580***	-6.589	-3.147***	-1.255***
	(0.830)	(0.483)	(1.014)	(6.310)	(1.018)	(0.319)
Dpos	-0.060	0.197	0.136	-1.236**	-1.067**	-0.037
	(0.282)	(0.396)	(0.141)	(0.490)	(0.530)	(0.263)
lnage	-0.093***	-0.093***	-0.094***	-0.083**	-0.081**	-0.082**
	(0.023)	(0.023)	(0.023)	(0.034)	(0.034)	(0.034)
empl2to49 (d)	0.055	0.056	0.059*	0.008	0.010	0.009
	(0.036)	(0.036)	(0.036)	(0.041)	(0.041)	(0.041)
empl50to250 (d)	0.011	0.012	0.015	-0.014	-0.011	-0.013
	(0.041)	(0.041)	(0.041)	(0.042)	(0.043)	(0.042)
listed (d)	0.119**	0.119**	0.117**	0.009	0.008	0.009
	(0.053)	(0.053)	(0.053)	(0.060)	(0.060)	(0.060)
foreign	-0.038	-0.039	-0.038	-0.014	-0.013	-0.012
	(0.033)	(0.033)	(0.033)	(0.044)	(0.043)	(0.043)
BankFin	-0.214***	-0.217***	-0.222***	-0.240***	-0.242***	-0.241***
	(0.041)	(0.040)	(0.038)	(0.048)	(0.048)	(0.048)
overdue (d)	0.132***	0.136***	0.141***	0.189***	0.190***	0.195***
	(0.045)	(0.045)	(0.045)	(0.054)	(0.053)	(0.054)
CredSale	-0.109***	-0.108***	-0.110***	-0.085***	-0.085***	-0.085***
	(0.023)	(0.023)	(0.023)	(0.032)	(0.032)	(0.033)
audit (d)	-0.045**	-0.044**	-0.042*	-0.019	-0.019	-0.016
	(0.021)	(0.022)	(0.022)	(0.031)	(0.030)	(0.031)
subsidies (d)	-0.104***	-0.104***	-0.103***	-0.098***	-0.097***	-0.099***
	(0.020)	(0.020)	(0.021)	(0.032)	(0.032)	(0.032)
arthro	0.484*	0.424	0.303	0.425	0.442*	0.291
Log likelihood	-1997.42	-2005.76	-2000.26	-876.51	-870.42	-871.24
No of obs.	2295	2295	2295	1075	1075	1075

Source: authors' calculations on BEEPs data. Note: Bivariate probit marginal effects: direct effects on R&D above and indirect effects via credit constraint below. Robust, sector clustered standard errors in parenthesis. Country, sector and year dummies included. ***, **, * stand for 1%, 5% and